

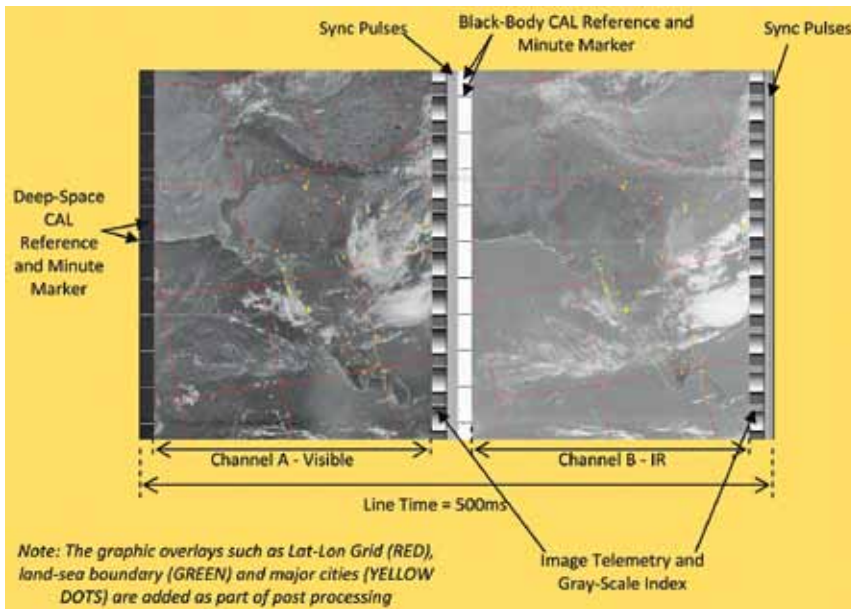


Fig. 1: The image components of the analogue APT

mitted digitally at a 665kbps rate with PSK modulation at main carrier frequencies of 1698MHz, 1708MHz, or 1702.5MHz. Therefore, the ground-receiving station becomes complex for an individual or amateur experimenter to construct. Each line has a pixel resolution of 1.1km and the HRPT image is transmitted using all the above five channels.

APT analogue transmission.

The automatic picture transmission (APT) protocol is the one we are currently using to receive images from the NOAA satellites. However, only two channels from the above five sensors are used for APT transmission, which are decided by the ground command-and-control of the satellite and thus are not user-selectable.

From the 10-bit ADC, the APT uses 8-bit MSB data to produce the 8-bit grey-scale levels for the analogue APT transmission. It uses a 2.4kHz audio tone and its amplitude is modulated as a function of the intensity level (grey-scale level) of the picture information. Also, the HRPT produces 360 lines per minute from which only two lines from the spectral channels are used for producing the APT data. Therefore,

the resulting line resolution gets reduced to 120 lines per minute.

Thus, with 120 lines per minute, we get two lines per second, or each line has a 0.5-second (500ms) duration. Let us now see how these 500ms get distributed between various signal components, as shown in Fig. 1.

1. The sync pulses. The sync pulses are transmitted in the form of square wave pulses for each set of the image component, that is, the Visible channel and the IR channel. Each set consists of seven pulses. For the Visible image channel, 1040Hz square wave pulses are used. While for the IR channel, 832Hz square wave pulses are used. These sync pulses appear as vertical lines in the primary raw image data received from the satellite and can be seen in Fig. 1 at the middle and right hand side of the image.

2. CAL references and time markers. The scanner uses two references while scanning the earth. It uses the Deep Space as a reference for the channel A visible imagery, and it appears as a vertical black strip at the beginning of the Visible channel, as can be seen at the left hand side of the image. Also, horizontal white

lines are overlaid for time reference as 1-Minute Marker.

For the channel B IR imagery, the Black-Body is used as the thermal reference and appears as vertical white strip at the beginning of the IR channel. Similarly, the time markers are overlaid as 1-Minute Marker but appear as black horizontal lines, as can be seen in the mid-right part of Fig. 1.

3. Telemetry data and grey-scale indices. For each of the Visible and IR channel, the image brightness data is transmitted as varying grey-scale levels and appear as horizontal strips called Wedges. It is used for temperature calibration of the image data in the respective image channel.

4. Channel A (Visible sensor) image. The channel A image section appears for 250ms of the scan-line from the Visible sensor of the satellite. Besides the CAL reference and the telemetry data, the major part contains the image data. During the earth scan, the visible light reflected from the earth's surface is used to construct this image. It therefore contains the oceans appearing as black, the clouds appearing as white, and the land as various levels of grey scale.

During night-time, this channel sends image data from another IR channel and has a clear and distinct image difference as compared to channel B's IR image data. The ground control decides the channel to be used during night-time pass of the satellite.

5. Channel B (IR sensor) image. Similarly, the channel B image section appears for the remaining of the 250ms and sends data from the IR sensor. Apart from the sync and telemetry data, the remainder of the time is used to send the sensor data from earth scan. Warmer objects appear as black or darker grey shades while colder objects appear white or lighter shades.

With the present scope of the